Field	Fill in
Team number	Team 5
Members	Nini Rizhamadze, Giulia Scarpa, Milena Moschetti, Thi Thuy Khuc, Lina-Sophie Reinsch, Hai Yen Bui
Question ID	Q15: What should the “you may also like” shelf show, and to whom?
Date	2026-10-20

1. PROBLEM & WHAT IT IS WORTH

- **The Question**: Which products should Lumen Beauty display on the "you may also like" recommendation shelf, and should these recommendations be defined by customer segment?
- **Stakeholder**: The brand founder and e-commerce merchandising team.
- **Problem Definition**: Lumen Beauty operates across multiple sales channels (livestream, search, social). Currently, the online checkout and product pages lack an optimized cross-sell recommendation engine, leading to missed opportunities to increase basket size and average order value (AOV).
- **Business Value**: Across 3,401 historical orders (**lb_order_lines.csv**), Lumen Beauty generated a total gross merchandise value (GMV) of CNY 1,173,482.47 with an average baseline of 1.45 items per order. Getting product discovery wrong results in inefficient shelf space utilization, lower basket density, and suppressed order values across 647 active purchasing customers. Identifying high-affinity product pairs ensures that digital recommendations directly drive cross-sell conversion without relying on generic or untargeted merchandising.

2. DATA

- **Datasets Used**:
  - **lb_order_lines.csv** (4,920 rows): The core transactional table containing *order_id, user_id, sku_id, sku_name, category, qty, unit_price, channel, stream_id, and order_date.*
  - **lb_users.csv** (1,000 rows): Customer profile data containing *user_id, city_tier, age_band, spend_total, n_orders, and high_value (*identifying top-tier customers, of which 130 are classified as *high-value).*
- **Data Cleaning & Preprocessing**:

- Excluded auxiliary tables (**lb_comments.csv, lb_follows.csv, lb_streams.csv)** to maintain a focus on order lines and purchasing behavior.
- Inspected for missing values, duplicate rows, and invalid unit prices or quantities; verified that all 647 unique purchasing users map cleanly to **lb_users.csv.**
- Grouped transactional rows by **order_id** into market baskets to evaluate co-purchasing frequencies.

3. METHOD & ASSUMPTIONS

- **Method Family**: Association Rule Mining (Market Basket Analysis, Week 3 framework).
- **Rationale**: Identifies probabilistic co-purchase relationships between items, directly determining which products naturally complement each other on a recommendation shelf.
- **Core Metrics & Thresholds**:
  - **Support**: Minimum threshold of $\geq 2\%$ to ensure product pairs appear frequently enough across orders to be statistically meaningful.
  - **Confidence**: Minimum threshold of $\geq 20\%$ to guarantee conditional purchase probability.
  - **Lift**: Threshold of > 1 to verify that item co-occurrence exceeds random chance.
- **Customer Segmentation**: Comparing association rules between high-value customers (high_value = 1; 130 users) and standard buyers to determine if premium segments require distinct recommendation bundles.
- **Assumptions & Limitations**: Association measures historical co-purchase correlation rather than direct causation, historical logs cannot observe past recommendation clicks or prove organic incremental sales without live experimentation.